Quaternionic Gate Fusion on S^3:
Geometry-Native Optimization for Single-Qubit Circuit Segments

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Abstract

This paper studies a local optimization pass for single-qubit circuit segments using the canonical quaternionic intermediate representation defined in a Canonical Quaternionic Intermediate Representation. A maximal run of adjacent single-qubit gates on one wire is translated into unit quaternions, multiplied in circuit order, re-normalized, and sign-canonicalized on the quotient $q \sim -q$ induced by global-phase removal. The result is a unique canonical fused representative in S^3 for the segment action modulo global phase. A layered reconstruction strategy then emits either no gate, an exact named gate, an exact axis-aligned rotation, or a generic single-qubit primitive when no simpler interpretable output is available. The paper proves exact equivalence preservation up to global phase, determinism of the fused representative under the fixed canonicalization rule, and minimality of the fully fused segment in the quaternionic IR itself. A small reference benchmark corpus shipped with the paper package is evaluated by an included plain-Python harness. On 13 measured segments containing 36 input single-qubit gates, the local fusion pass reconstructs to 11 output gates while preserving exact segment action under the authoritative canonical-quaternion equivalence check. Two segments eliminate to identity, and 8 of 13 outputs are recovered as either named gates or axis-aligned rotations. These measurements are limited to the included single-qubit reference corpus and are not presented as a claim of universal compiler superiority.

Keywords: quaternionic gate fusion; single-qubit optimization; compiler method; SU(2); unit quaternions; local circuit simplification

1 Introduction

Foundations of Quaternionic Quantum Computing established the fixed quaternion–SU(2) convention used throughout the repository. A Canonical Quaternionic Intermediate Representation, defined a canonical single-qubit intermediate representation $\mathrm{u1q}(w, x, y, z)$ together with deterministic normalization and sign-canonicalization rules. The present paper asks the next compiler-facing question: once the representation exists, what optimization rule does it induce?

The answer is local and operational. If a circuit contains a maximal run of adjacent single-qubit gates on one wire, then those gates may be fused by quaternion multiplication. This is not an analogy or a heuristic. It is the group law of the representation chosen in papers 1 and 2. The optimizer maps the segment into quaternions, multiplies in circuit order, re-canonicalizes the product, and then reconstructs an interpretable output whenever possible.

This paper is therefore not a manifesto about quaternions. It is a method paper about a local optimization pass. The pass is limited to single-qubit segments between barriers such as entangling

gates, measurement, reset, or classical control boundaries. It does not optimize entangling gates, and it does not claim to solve full-circuit synthesis or compiler optimality.

Three contributions are the focus:

(1) a formally specified fusion operator on single-qubit segments;

(2) a reconstruction layer that is exact when identity, named-gate, or axis-aligned forms are available, but otherwise falls back honestly to a generic one-qubit primitive; and

(3) a measured evaluation on a shipped reference corpus with a clearly stated syntax-preserving baseline.

2 Background and dependency on papers 1–2

2.1 The inherited mathematical model

We adopt without change the convention fixed in paper 1:

$$\Phi(a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}) = \begin{pmatrix} a - di & -c - bi \\ c - bi & a + di \end{pmatrix} = aI - i(b\sigma_x + c\sigma_y + d\sigma_z). \tag{1}$$

Under this convention, unit quaternions model SU(2) exactly, and single-qubit gates in U(2) admit representatives in SU(2) after removal of global phase.

Proposition 2.1 (Single-qubit gate actions admit unit-quaternion representatives)**.** *For every single-qubit gate $U \in \mathrm{U}(2)$ there exists a unit quaternion $q \in \mathbb{H}$ such that $\Phi(q)$ represents the same single-qubit action as U on pure states modulo global phase. The representative q is unique up to sign.*

Proof. By paper 1, every $U \in \mathrm{U}(2)$ admits a representative $V \in \mathrm{SU}(2)$ after removing determinant phase, and V is unique up to sign. The restriction of Φ to the unit quaternions is an isomorphism onto SU(2), so V corresponds to a unique sign pair $\{q, -q\}$. □

2.2 The inherited canonical IR

Paper 2 defined the canonical single-qubit IR object

$$\mathrm{u1q}(w, x, y, z),$$

interpreted as the unit quaternion

$$q = w + x\mathbf{i} + y\mathbf{j} + z\mathbf{k}, \qquad w^2 + x^2 + y^2 + z^2 = 1.$$

It also fixed the canonicalization map

$$\mathrm{Can} = \mathrm{SignCan} \circ \mathrm{Norm}$$

with primary rule $w > 0$ and an equatorial tie-break when $w = 0$.

Remark 2.2. The present paper depends conceptually on paper 2 for determinism. Quaternion multiplication alone would not choose a unique stored representative, because q and $-q$ encode the same single-qubit action modulo global phase. Fusion therefore uses multiplication *followed by* canonicalization, not raw multiplication alone.

3 Problem setup

Definition 3.1 (Single-qubit segment)**.** *A* single-qubit segment *on wire a is a maximal consecutive run of gates that act only on wire a, bounded by the beginning or end of the circuit, or by a boundary operation that lies outside the present single-qubit model for that wire. Typical boundaries include entangling gates, measurement, reset, and classically controlled branching points.*

Definition 3.2 (Target optimization problem)**.** *Given a single-qubit segment*

$$\mathrm{seg} = (g_1, g_2, \ldots, g_n),$$

find a replacement segment seg' *such that:*

(a) seg' *represents exactly the same single-qubit action as* seg *modulo global phase;*

(b) seg' *is deterministic under the fixed canonicalization convention; and*

(c) $\mathrm{G}(\mathrm{seg}')$ *and* $\mathrm{D}(\mathrm{seg}')$ *are no larger than their baseline values for the original segment.*

Remark 3.3. For a purely sequential single-qubit segment, gate count and segment-local depth coincide. In a larger circuit, however, this paper claims reduction only for the single-qubit run itself. Entangling depth is not changed by the present method.

Definition 3.4 (Authoritative equivalence criterion)**.** *For two single-qubit segments* seg *and* seg'*, define*

$$\mathrm{Eq}(\mathrm{seg}, \mathrm{seg}')$$

to hold when their canonical fused quaternions agree exactly:

$$\mathrm{Can}(q_n \cdots q_1) = \mathrm{Can}(q'_m \cdots q'_1),$$

where the q_i and q'_j are the translated unit-quaternion representatives of the gates in application order.

4 Quaternionic fusion method

4.1 Translation and multiplication order

Let

$$\mathrm{seg} = (g_1, g_2, \ldots, g_n)$$

be a single-qubit segment written in circuit application order. For each gate g_k, let q_k be its canonical unit-quaternion representative under the map of paper 2.

Definition 4.1 (Quaternionic fusion operator)**.** *Define the fused quaternionic payload of the segment by*

$$\mathrm{Fuse}(\mathrm{seg}) = \mathrm{Can}(q_n q_{n-1} \cdots q_1).$$

Remark 4.2 (Order convention). Equation (4.1) is intentionally right-to-left in the gate index. The first gate applied to the state is g_1, and under (1) one has

$$\Phi(q_n q_{n-1} \cdots q_1) = \Phi(q_n)\Phi(q_{n-1}) \cdots \Phi(q_1),$$

which is the matrix product for sequential application in that order.

4.2 Normalization and sign-canonicalization

The fusion pass uses the canonicalization rule inherited from paper 2:

(1) translate each gate into a canonical unit quaternion;

(2) multiply in circuit order as in (4.1);

(3) re-normalize if numerical drift has occurred; and

(4) apply sign-canonicalization with the primary rule $w > 0$ and the equatorial tie-break when $w = 0$.

Definition 4.3 (Baseline flow)**.** *The benchmark baseline used in this paper is the* syntax-preserving baseline

$$\mathrm{Base}(\mathrm{seg}) = \mathrm{seg},$$

meaning that no fusion is performed and the gate list is left unchanged.

5 Axis-aware reconstruction

The fused quaternion is a complete canonical segment representative, but a compiler often benefits from an interpretable output layer.

Definition 5.1 (Axis-aware reconstruction)**.** *Given a canonical fused quaternion q, the reconstruction operator*

$$\mathrm{Recon}(q)$$

applies the following layered rule:

(1) if $q = (1, 0, 0, 0)$, emit no gate;

(2) if q matches a stored exact canonical tuple for a named gate such as X, Y, Z, H, S, $S^\dagger$, T, or $T^\dagger$, emit that named gate;

(3) if exactly one of (x, y, z) is nonzero, recover the corresponding exact axis-aligned rotation $R_x(\theta)$, $R_y(\theta)$, or $R_z(\theta)$;

(4) otherwise emit a generic canonical one-qubit primitive $\mathrm{u1q}(w, x, y, z)$ *or an equivalent internal opaque primitive.*

Remark 5.2. The reconstruction layer is not the equivalence proof. Equivalence is certified by canonical quaternion equality. Reconstruction is a presentation and compiler interface layer on top of that exact representation.

Proposition 5.3 (Exact axis-aligned cases admit direct recovery)**.** *If the canonical fused quaternion has exactly one nonzero imaginary component, then it reconstructs exactly to an axis-aligned rotation family member $R_x(\theta)$, $R_y(\theta)$, or $R_z(\theta)$.*

Proof. If only one of (x, y, z) is nonzero, then the quaternion already has axis–angle form about a coordinate axis:

$$q = \cos(\theta/2) + \sin(\theta/2)\mathbf{i}, \quad q = \cos(\theta/2) + \sin(\theta/2)\mathbf{j}, \quad \text{or} \quad q = \cos(\theta/2) + \sin(\theta/2)\mathbf{k}.$$

Under (1), these are exactly $R_x(\theta)$, $R_y(\theta)$, and $R_z(\theta)$. □

Remark 5.4 (Named-gate recovery is not globally unique). The canonical fused quaternion is unique under the fixed convention. A human-readable named-gate decomposition generally is not. The present paper only claims uniqueness for the quaternionic representative and exact recovery for the special cases explicitly recognized by Recon.

6 Theoretical properties

Proposition 6.1 (Fusion preserves exact segment action up to global phase)**.** *For every single-qubit segment* seg*,*

$$\mathrm{Eq}(\mathrm{seg}, \mathrm{Fuse}(\mathrm{seg}))$$

holds when Fuse(seg) *is interpreted as its fused canonical representative.*

Proof. Let $q_1, \ldots, q_n$ be the gate representatives in application order. Under (1), the matrix product of the segment is

$$\Phi(q_n) \cdots \Phi(q_1) = \Phi(q_n \cdots q_1).$$

Canonicalization does not change the represented segment action modulo global phase, because it only normalizes and chooses the unique representative from the sign pair $\{q, -q\}$. Hence the fused canonical quaternion represents exactly the same single-qubit action as the original segment. □

Proposition 6.2 (Determinism of the fused representative)**.** *For every nonempty single-qubit segment* seg*, the value* Fuse(seg) *is uniquely determined by the segment action and the fixed canonicalization convention of paper 2.*

Proof. The translated product $q_n \cdots q_1$ is uniquely determined by the segment action up to the sign ambiguity inherited from SU(2). The map Can chooses exactly one representative from each sign class. Therefore the fused representative is unique. □

Corollary 6.3 (Segment replacement by one canonical payload)**.** *Every nonidentity single-qubit segment may be replaced by one canonical quaternionic payload without changing the represented segment action modulo global phase. Identity segments may be replaced by the empty segment.*

Proof. This follows immediately from the previous two propositions. □

Proposition 6.4 (Minimality in the quaternionic IR)**.** *Within the quaternionic IR itself, the fully fused representation of a nonidentity single-qubit segment has minimal gate count equal to one.*

Proof. The quaternionic IR stores segment action as a single canonical payload $\mathrm{u1q}(w, x, y, z)$. Any nonidentity segment action has at least one such payload, and no smaller nonempty representation exists inside that IR. Identity is the unique case in which the representation can be reduced to zero emitted gates. □

7 Benchmark methodology

7.1 Reference corpus

The benchmark corpus shipped with this paper package is deliberately modest. It is intended as a reference evaluation of the local fusion method rather than a claim of production-scale compiler performance. The included harness evaluates 13 segments across five categories:

(1) textbook single-qubit chains;

(2) rotation-heavy segments;

(3) mixed named-gate segments;

(4) single-qubit runs extracted from small two-qubit circuits by entangling boundaries; and

(5) an ansatz-style single-wire segment.

7.2 Metrics

For each segment seg, the harness records:

$$\begin{aligned} \mathrm{G}_{\mathrm{orig}}(\mathrm{seg}) &= \text{original number of gates}, \\ \mathrm{G}_{\mathrm{opt}}(\mathrm{seg}) &= \text{number of reconstructed output gates}, \\ \mathrm{D}_{\mathrm{orig}}(\mathrm{seg}) &= \text{original segment-local depth}, \\ \mathrm{D}_{\mathrm{opt}}(\mathrm{seg}) &= \text{optimized segment-local depth}. \end{aligned}$$

It also records:

- whether the reconstructed output is identity, named, axis-aligned, or generic;
- whether the reconstructed output passes the authoritative equivalence check after re-translation into canonical quaternion form; and
- whether the segment collapsed to one canonical representative before reconstruction.

7.3 Baseline

The baseline is the syntax-preserving flow Base. It performs no fusion and leaves the segment unchanged. The comparison is therefore intentionally modest: the paper measures what the local quaternionic pass does relative to doing nothing on those segments.

Remark 7.1. No comparison is claimed here against a major external compiler stack. The present goal is to study a local method and to ship a reproducible reference harness for it.

8 Worked examples and measured reference-corpus results

8.1 Worked examples

Example 8.1 ($R_x(\pi/2); R_x(\pi/2)$). The translated quaternions are

$$q_1 = q_2 = \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0, 0 \right).$$

Hence

$$\mathrm{Fuse}(\mathrm{seg}) = \mathrm{Can}(q_2 q_1) = (0, 1, 0, 0).$$

Axis-aware recovery first identifies the x axis, yielding $R_x(\pi)$, and the named-gate table can further recognize that canonical tuple as X. The segment changes from gate count and local depth 2 to 1.

Example 8.2 ($R_z(\pi/4); R_z(\pi/4)$). The fused quaternion is

$$\mathrm{Can}\left(\left(\cos\frac{\pi}{8}, 0, 0, \sin\frac{\pi}{8}\right)^2\right) = \left(\frac{1}{\sqrt{2}}, 0, 0, \frac{1}{\sqrt{2}}\right).$$

This is axis-aligned about z and is also recognized as the named gate S. Again the segment changes from $(2, 2)$ in count and local depth to $(1, 1)$.

Example 8.3 ($H; Z; H$). Using the canonical representatives of paper 2,

$$q_H = \left(0, \frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right), \qquad q_Z = (0, 0, 0, 1).$$

Fusion gives

$$\mathrm{Fuse}(\mathrm{seg}) = \mathrm{Can}(q_H q_Z q_H) = (0, 1, 0, 0),$$

which reconstructs to X. This is the expected similarity transformation, now expressed directly as a quaternionic local optimization.

Example 8.4 (A segment between two entangling boundaries). Consider the wire-0 run

$$T^\dagger; R_x(\pi/4)$$

extracted from a larger two-qubit circuit between two CNOT barriers. Fusion gives

$$\mathrm{Fuse}(\mathrm{seg}) \approx (0.8535534,\ 0.3535534,\ 0.1464466,\ -0.3535534).$$

This output is not axis-aligned and does not match the exact named-gate lookup table, so reconstruction emits a generic u1q primitive. Even in this case, the local optimization still reduces the segment from two gates to one while preserving exact segment action modulo global phase.

8.2 Measured reference-corpus results

The included benchmark harness in `artifacts/code/benchmark_gate_fusion.py` computes the measured results stored in `artifacts/code/benchmark_results.json`. On the shipped reference corpus:

- 13 segments were evaluated;
- original gate count was 36 and optimized gate count was 11;
- original segment-local depth was 36 and optimized segment-local depth was 11;
- 2 segments collapsed to identity and emitted no gate;
- all 13 segments collapsed to a single canonical quaternionic representative before reconstruction; and
- all 13 equivalence checks passed.

These numbers correspond to a gate-count and segment-depth reduction of

$$36 - 11 = 25,$$

or approximately 69.4% relative to the syntax-preserving baseline on this reference corpus.

The measured reconstruction breakdown is:

Reconstruction kind	Count
named gate (including identity)	6
axis-aligned rotation	2
generic one-qubit primitive	5

Thus 8 of 13 outputs (about 61.5%) are recoverable as named or exact axis-aligned forms, while the remaining 5 are emitted as generic canonical primitives.

Category-level results are shown in Table 1.

Category	Segments	Original gates	Optimized gates	Reduction
Textbook single-qubit chains	3	7	3	4
Mixed named-gate segments	3	9	2	7
Rotation-heavy segments	4	11	3	8
Segments from two-qubit circuits	2	4	2	2
Ansatz-style segments	1	5	1	4
Total	13	36	11	25

Table 1: Measured gate-count results on the shipped reference corpus under the syntax-preserving baseline. Because every benchmark item is a sequential single-qubit segment, the same totals also describe segment-local depth.

Remark 8.5. These measurements are intentionally narrow. They demonstrate that the local fusion rule is operational and measurable on a shipped corpus. They do not establish universal superiority over other compiler flows, and they do not address whole-circuit optimality.

9 Standard mathematics vs. RQM Technologies framing

9.1 Standard ingredients

The following ingredients are standard:

(1) single-qubit gates live in U(2) and admit SU(2) representatives modulo global phase;

(2) unit quaternions model SU(2) under the fixed convention (1);

(3) sequential single-qubit gate composition corresponds to quaternion multiplication; and

(4) $q \sim -q$ is the residual ambiguity after removal of global phase.

9.2 Project-specific contribution

The project-specific contribution of this paper is the optimization method:

(a) use the canonical u1q representation of paper 2 as the working IR;

(b) define a local fusion pass by quaternion multiplication plus re-canonicalization;

(c) use a layered reconstruction rule for identity, named-gate, axis-aligned, and generic outputs; and

(d) evaluate the pass against a syntax-preserving baseline on a shipped reference corpus.

These are compiler-method choices. They are not new mathematics, and they are not claims about hardware advantage or universal circuit optimality.

10 Scope, limitations, and non-claims

This paper does *not*:

(1) optimize entangling gates;

(2) define a full quantum compiler;

(3) prove universal superiority over all baseline compilers;

(4) claim hardware speedups;

(5) claim overall circuit optimality; or

(6) solve the general single-qubit synthesis problem in arbitrary gate sets.

It defines and studies a local optimization pass for single-qubit segments.

11 Conclusion

The quaternionic model is operationally useful because it turns adjacent single-qubit fusion into ordinary multiplication in a canonical representation. Once paper 2's deterministic u1q representation is in place, paper 3's fusion rule follows naturally:

$$\mathrm{Fuse}(\mathrm{seg}) = \mathrm{Can}(q_n \cdots q_1).$$

That rule preserves exact segment action modulo global phase, yields a unique canonical fused form, and supports an honest reconstruction layer that can emit identity, named gates, axis-aligned rotations, or generic primitives as needed.

On the shipped reference corpus, the local pass reduces 36 input single-qubit gates to 11 reconstructed outputs while preserving exact equivalence under the authoritative canonical-quaternion check. This is not a claim of universal compiler optimality. It is a concrete result about one local method: a geometry-native optimization pass for single-qubit segments.

A Quaternion product formula

For

$$q_1 = (w_1, x_1, y_1, z_1), \qquad q_2 = (w_2, x_2, y_2, z_2),$$

the product used by the fusion pass is

$$q_2 q_1 = \big(w_2 w_1 - x_2 x_1 - y_2 y_1 - z_2 z_1,\, w_2 x_1 + x_2 w_1 + y_2 z_1 - z_2 y_1,$$

$$w_2 y_1 - x_2 z_1 + y_2 w_1 + z_2 x_1,\, w_2 z_1 + x_2 y_1 - y_2 x_1 + z_2 w_1\big).$$

B Benchmark harness notes

The reference harness included with this paper package uses only the Python standard library. It translates each benchmark segment into quaternions, fuses it by (4.1), applies reconstruction, and verifies equivalence by re-translating the reconstructed output and comparing canonical quaternions.

C Companion figures and code

Publication-oriented SVG figures are provided in `artifacts/figures/`. The benchmark harness and measured results are provided in `artifacts/code/benchmark_gate_fusion.py` and `artifacts/code/benchmark_results.json`.